

Malek Bsaissa

Cloud, DevOps & Software Engineering

Ariana, Tunisia | dev.malekbsaissa@gmail.com
malekbsaissa.vercel.app | github.com/AtfastrSlushyMaker
linkedin.com/in/malek-bsaissa-8861b229b

Final-year engineering student seeking an end-of-study internship. Experience building full-stack applications, automating cloud infrastructure, and integrating AI services.

Education

ESPRIT School of Engineering | 2022 - 2027 (expected)

- Engineering Degree in Computer Science. Fifth and final year; Cloud Computing and DevOps specialization.

Technical Skills

Cloud & DevOps

- OpenStack, Kubernetes, Azure AKS, Docker, Ansible, Terraform, Karmada, GitHub Actions, Linux, NGINX, Prometheus, Grafana.

Software Engineering

- Java, Python, TypeScript, JavaScript, PHP, SQL; Spring Boot, Angular, React, FastAPI, Laravel, Symfony, JavaFX; PostgreSQL, MySQL, REST APIs, WebSocket, Git.

AI & Data

- LLM integration, tool calling, retrieval-grounded answers, pandas, scikit-learn, LightGBM, CesiumJS.

Experience

Banque de Tunisie | Full-Stack & Cloud Engineering Intern | Jul - Aug 2026

- Built MEMO with Angular, Spring Boot and PostgreSQL: tenant-scoped workplace operations, messaging, audit trails and a separate platform console.
- Integrated a FastAPI assistant with streamed, source-grounded answers and delegated authorization enforced by the Spring backend.
- Deployed an AKS demonstration environment using Terraform and GitHub Actions OIDC; added container builds and Prometheus/Grafana monitoring.

Smart Skills, Ariana | Full-Stack Developer Intern | Jun - Aug 2025

- Built MySkills, a training management platform for administrators, coordinators, trainers and trainees using Laravel, React and MySQL.
- Implemented REST APIs, Laravel Sanctum authentication, role-based access control, registration and approval workflows.
- Added automated certificates, in-app and email notifications, and reporting.

Tunisair | Operations Intern | Jun 2023

- Supported and observed ticketing, refund processing and daily operations at Tunis-Carthage International Airport.

Selected Projects

Hybrid Cloud Infrastructure | Academic team project | 2026

- Automated OpenStack infrastructure and Kubernetes configuration with Ansible, Heat and kubeadm; connected private-cloud workloads to Azure AKS through Karmada.
- Worked with Nova, Neutron, Keystone, Glance, Heat, Horizon and Octavia, with Cinder block storage and Swift object storage in the infrastructure. The final application used local-path MySQL volumes.
- Configured six propagation policies and demonstrated four application replicas split across two clusters; added Prometheus and Grafana monitoring.

Elif & Elif AI Agent | Community module and companion service | 2025 - 2026

- Built posts, threaded discussions, voting, moderation, mentions, notifications and real-time chat using Angular, Spring Boot, MySQL and WebSocket.
- Created a separate FastAPI/Groq service for natural-language community search, bounded tool calls and answers grounded in retrieved community context.

AtlasMesh | Solo geospatial project | 2026

- Combined 18 live data sources on a React/CesiumJS globe with a Node.js and TypeScript backend.
- Implemented real-time WebSocket updates, satellite propagation, entity search and independently managed map layers.

The 12th Player | Team machine-learning project | 2025 - 2026

- Contributed to football analytics served through FastAPI and React: KNN standings, Random Forest match outcomes, KMeans team styles and LightGBM talent scouting.

WamiaGo | Bicycle module, team project | 2024

- Built electric-bicycle rental workflows, station availability and QR-code unlocking across JavaFX desktop and Symfony web applications with shared MySQL data.

Ripple Insurance System | Academic project

- Worked on accident-detection simulation using C++, Qt and Arduino sensor integration with automated alerts.

Languages

Arabic: native | English: approximately B2 | French: B1

- Language levels are self-assessed.